Partition-Tolerant Federation Across Emulated ISP Islands

Anonymous submission

Abstract

An Internet shutdown is rarely a clean power switch. Transit to the outside world can be withdrawn while domestic networks keep carrying packets among themselves, leaving reachable fragments that recent measurement work calls connectivity islands. We ask a deliberately narrow question: when some domestic IP paths survive, can independently operated forum servers keep accepting signed posts and keep reconciling them across whatever graph remains? Our prototype combines four mechanisms that are individually familiar and, to our knowledge, have not been composed for this operating regime: canonical protobuf envelopes admitted as the exact bytes their author signed, operator receipts bound to those bytes, caller-initiated synchronisation that survives asymmetric reachability, and an explicitly trusted dual-homed relay between islands. On a four-node testbed whose shared exchange segment is withdrawn mid-run, a dependent certificate, community, and post crossed the severed boundary in 0.5, 1.2, and 2.1 seconds—within a factor of two of the same chain measured while the exchange was intact. An eight-node chain delivered 200 of 200 envelopes seven hops with a 4.125 s median and a 6.115 s 99th percentile, and per-hop cost tracked a queueing model to within 89 ms across a fourfold change in the drain interval. TypeScript, Rust, and Python implementations agreed on all 16 canonical vectors. A signature-invalid flood produced no database writes, though a rate limiter shed 90%

of it before the pipeline was reached, which we report rather than headline. Against a median gzipped ActivityPub activity our envelopes are 4.4–6.9× smaller, at the cost of Fediverse interoperability and JSON-LD extensibility. These results establish partition tolerance under surviving IP paths on a single-host testbed. They establish nothing about total blackouts, real inter-provider deployment, or resourced denial of service.

Index Terms

federated systems, Internet shutdowns, network partitions, signed objects, store-and-forward

I. Introduction

Treating an Internet shutdown as a binary event obscures what operators actually do. Disruptions arrive as throttling, destination blocking, protocol filtering, or the withdrawal of international transit, and they frequently leave domestic reachability partly intact [1], [2]. Baltra *et al.* formalised the resulting structures, naming networks severed from the Internet core *islands* and persistent partial reachability *peninsulas* [3]. Their contribution is measurement: they detect and name the condition. Ours is a system that has to keep working inside it. The question we pose is correspondingly narrow. If domestic IP paths survive a transit cut, can independent forum servers continue to admit signed posts and reconcile them across the surviving graph?

Adjacent systems answer adjacent questions. Circumvention tools such as Telex and Snowflake route a user to an external service through cooperating infrastructure or ephemeral proxies [4], [5]; both presume that some path out of the country exists, which is precisely what a transit cut removes. Device-to-device systems—Anix, Rangzen, Moby, Twimight—abandon the packet network entirely and carry messages on Bluetooth or Wi-Fi Direct encounters [6]–[9], which is the right approach for a total blackout and costs latency, bandwidth, and battery

to take. Neither family specifies how independently operated *domestic servers* should validate, store, and relay public forum objects while partial IP connectivity persists. That gap is what this paper occupies.

Our design rests on a single discipline: an author signs one exact byte sequence, and every server that later handles those bytes verifies them unchanged. No relay may normalise, repair, or re-encode an object on its way through. Acceptance produces a receipt signed over the same bytes, so an operator's acknowledgement is bound to what was actually admitted rather than to an identifier the operator chose. Synchronisation is always initiated by the caller, so a node with no reachable inbound address still both publishes and recovers. When two islands cannot reach each other at all, an explicitly trusted dual-homed server validates, stores, and relays between them, running the same admission code as every other node rather than forwarding packets.

We claim four contributions:

- a byte-preserving admission path applied identically to locally submitted and federated objects, which removes the failure mode in which a relay repairs a malformed object into a verifying one;
- caller-initiated delivery, streaming, and backfill, giving bidirectional federation to nodes that can open connections but not accept them;
- a quota-bounded relay policy that revalidates every object it forwards and excludes the peer it arrived from, so trust governs rate rather than validity; and
- measurements from a testbed in which the inter-island path is withdrawn mid-run, together with cross-language byte-agreement vectors, a hop-scaling chain, a signature-invalid flood, and a compression-aware ActivityPub size comparison.

We claim *partition tolerance under surviving IP paths* and

nothing broader. The design cannot manufacture a route through a total communication blackout, cannot conceal a visible relay from the network operator hosting it, and cannot compel an operator to accept a valid post.

II. Background and Research Gap

Federated and signed systems. ActivityPub delivers social activities to server inboxes [10]; Matrix and the AT Protocol federate rooms and replicate repositories [11], [12]; Nostr and Secure Scuttlebutt replicate author-signed content [13], [14]. Each supplies a mechanism we reuse. None assumes that the server graph itself may fragment: reachable peer addresses remain a background assumption, and no member of the family combines exact-byte admission, operator receipts, outbound-only recovery, and an explicit two-island relay policy in the arrangement evaluated here.

Blackout communication. Anix, Rangzen, Moby, and Twimight move data through physical proximity [6]–[9]; Dolphin smuggles data through a cellular voice channel when packet service is denied [15]. These designs trade latency, throughput, and energy for independence from any domestic server path. Ours makes the opposite trade and inherits the mirror-image weakness: it is fast while servers remain mutually reachable and it stops entirely when every IP path is gone. The two families are complements, not competitors, and we make no throughput or latency comparison against them because the workloads are not commensurable.

Accountability. Certificate Transparency established append-only logs, signed tree heads, and inclusion proofs as a way of making certain misbehaviour visible after the fact [16], [17]; PeerReview and CoSi pursue stronger notions through witnesses and cosigning [18], [19]. Our receipts are a deliberately weaker instrument in that lineage: a receipt proves that one operator acknowledged one specific byte string at one moment, not that

the operator accepts all valid objects, forwards them promptly, or refrains from silently declining the next one.

The gap sits between normally routed federation, which assumes the server graph holds, and device-to-device networks, which assume it has vanished. We assume instead that servers stay powered and at least one domestic path survives, and the objective under that assumption is to admit an authored object once, preserve its signed bytes without alteration, and let every reachable replica reach the same verdict on its validity.

III. System Design

A. Architecture

We designed the system to be censorship-resistant, pseudonymous, and unstoppable regardless of internet connectivity. Connectivity degradation is modeled as follows: L0 – normal global reach/Tor connectivity; L1 – national partition (international transit cut but domestic IX alive); L2 – ISP island (inter-ISP peering cut); L3 – bridged islands via a dual-homed relay; and L4 – total isolation, overcome by the Reticulum network stack. Ingress is therefore heterogeneous and multi-protocol to support these levels: ordinary HTTP/gRPC over surviving IP, Tor v3 onion services when an external Tor path persists, and an optional Reticulum/LoRa sidecar for small high-priority messages when IP is unavailable (BULK rejected, disabled by default). This paper evaluates only the IP-networking part (HTTP for clients, gRPC for federation) and focuses on L0–L3. Whatever the ingress path, every object on the wire is one uniform canonical signed packet traversing the same 19-stage pipeline; the first 12 stages perform admission with no persistent writes. After commit, the server returns a signed receipt and enqueues federation work in a durable outbox, which a scheduler drains by priority class and then grouped by (destination, source uplink, traffic class). A bridge is an ordinary server implementing that

same stack, dual-homed onto two islands — not a transparent packet forwarder — and the distinction matters: everything it relays, it has first independently validated, stored, and quota-bounded under its own key. Parallel to the servers, independent audit-log services receive a public audit certificate only after the server has accepted the write; they verify the certificate against the canonical envelope and the receipt, then append it to a separate hash-chained record, so that an acknowledgement a server later hides remains reproducible from outside. The complete system, including the reachability ladder that motivates the design, the on-device signing boundary, the canonical envelope, the unified admission pipeline, the per-node persistence stack, and the independent audit-log services, is shown in Fig. 1.

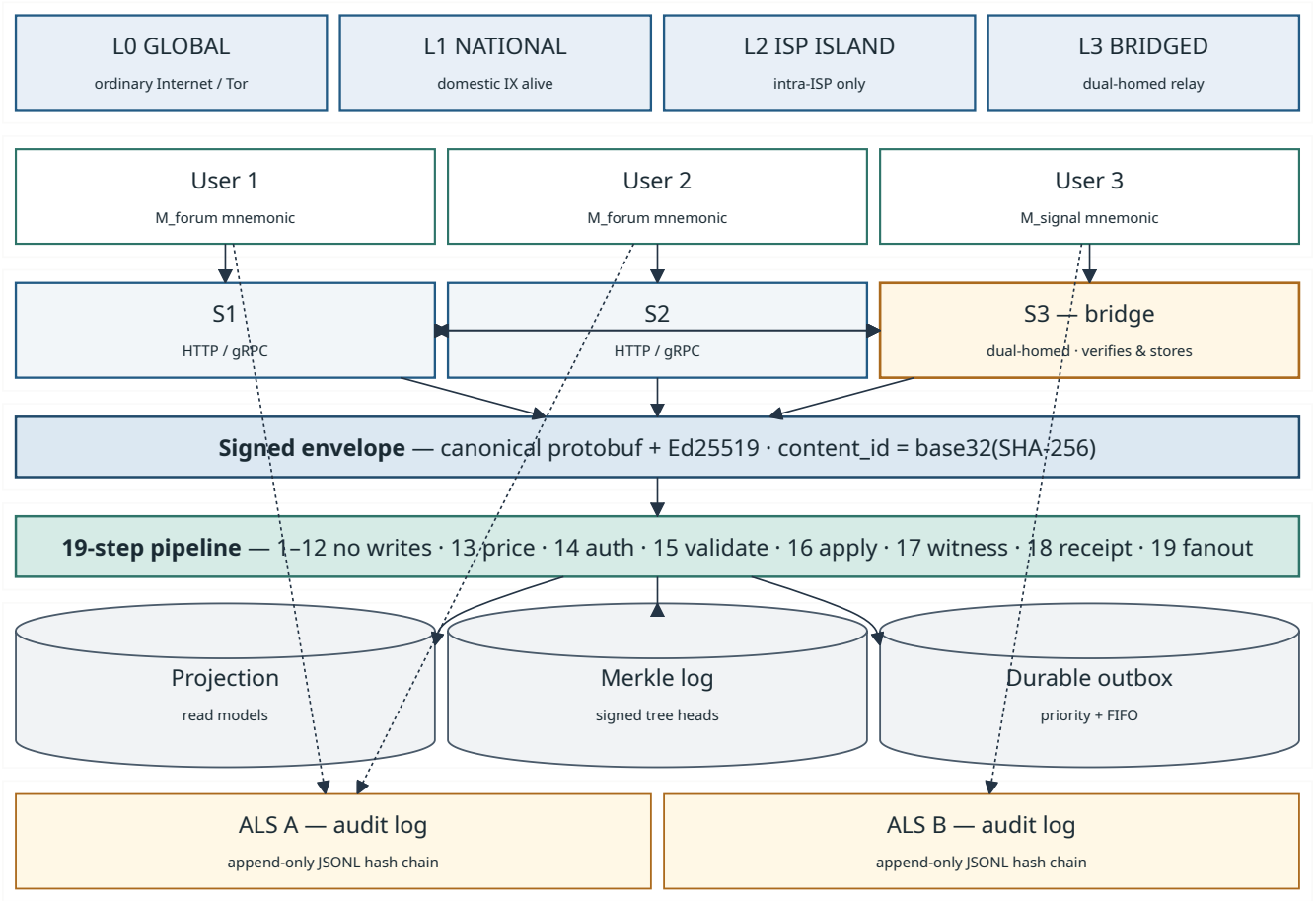

Fig. 1. System architecture. Clients sign canonical envelopes on-device and submit them to forum nodes over HTTPS or gRPC; every node runs the same 19-step admission pipeline, persists accepted envelopes into projections and a Merkle log, and exchanges exact signed bytes with peers via Deliver / StreamActivities / Backfill. Independent audit-log servers (ALS A, ALS B) receive public audit certificates after acceptance. The dual-homed node S3 acts as a bridge under the same admission policy. The reachability scope at the top indicates which paths are usable at a given moment.

B. Signed objects and admission

An envelope carries a domain tag, an object type, a payload, a public key, a nonce, and an expiry. Its identifier is the SHA-256 digest of a specified canonical byte string, and an Ed25519 signature covers that same string [20]. Because protobuf does

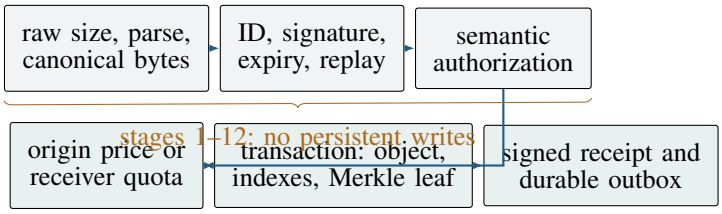

Fig. 2. Admission pipeline. State changes begin only after the cheap byte and validity checks have all succeeded.

not guarantee a unique serialisation [21], we constrain a profile on top of it: ascending field order, minimal varints, no unknown fields retained, and—the operative rule—a decode followed by a re-encode that must reproduce the received bytes exactly. An object that fails that comparison is rejected rather than repaired.

Strict-decode-then-verify-what-arrived is long-established practice in DER, JWS, and CT's leaf format; what it buys here is specific to relaying. A federation node that decoded a peer's bytes and re-encoded them before verification would silently normalise a malformed object into a well-formed one and then find that it verifies. Under our rule the receiver never normalises attacker-controlled bytes before checking the signature, so no relay can launder an object into validity.

Fig. 2 summarises the admission path. Stages 1 through 12 cover size, parsing, canonicality, domain separation, identifier derivation, signature, expiry, replay, and authorisation, and none of them writes to persistent storage—an invalid flood therefore cannot amplify into write load, a property we measure in §V. Later stages charge origin pricing or a receiver quota, commit the raw object together with its derived indexes in a single transaction, append its hash to a Merkle log, and only then enqueue federation. A uniqueness constraint in the database, rather than a read-then-write check, makes exact retries idempotent under concurrency.

On acceptance the server signs a receipt binding the object

identifier, its leaf position, the resulting tree head, and an inclusion proof, so the acknowledgement commits to the bytes admitted rather than to a label the operator is free to change afterwards. Peers additionally exchange signed tree heads. Two heads of equal size with differing roots are an unambiguous conflict; a larger head is recorded as pending until a consistency proof can be fetched. We are explicit about the reach of this rule. It is stale-head hygiene that eliminates a class of self-inflicted false positives—and it does catch outright tree shrinkage—but an adversary who rewrites history while keeping the tree non-decreasing passes both clauses undetected, and equivocation between two peers cannot be caught by any purely local comparison. Detecting either requires consistency proofs and cross-observer exchange, which the partition we are designing for is precisely what obstructs.

C. Partition-aware federation

Three remote procedure calls, all initiated by the caller, cover ordinary delivery and recovery. DELIVER pushes a batch, STREAMACTIVITIES resumes from a durable cursor, and BACKFILL requests objects a node knows it is missing. Because none of the three requires the calling node to accept an inbound connection, a server behind carrier-grade network address translation participates fully in both directions over connections it opens itself. This is reachability asymmetry, not discovery: at least one peer address must already be known.

The scheduler drains independent per-peer queues concurrently under a bounded lease, with deadlines, exponential backoff, and idempotency keys. That structure exists because its absence was a real defect, described in §V. One caveat is worth stating in the design rather than buried in limitations: the deadline bounds how long a drain pass waits on a peer, not how long the underlying transport keeps trying, because the sender takes

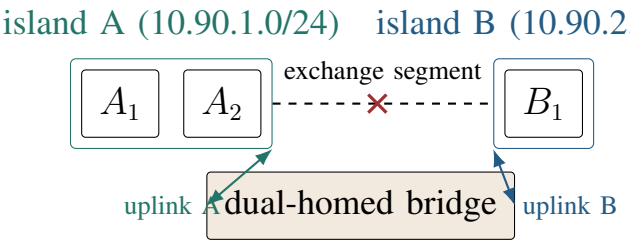

Fig. 3. Route-cut topology. Four application nodes span two isolated network segments joined by a shared exchange, which the experiment withdraws mid-run. The bridge is an application server, not an IP forwarder.

no cancellation token. A stuck call can therefore hold resources past its deadline even though it no longer blocks anyone.

Fig. 3 shows the route-cut topology. Two ordinary nodes sit in island A and one in island B; a shared exchange segment connects them, standing in for an Internet exchange point. The bridge holds one address on each island network. Once the exchange is withdrawn, both islands must already have configured trust in the bridge—trust that governs the rate and priority a peer's traffic receives, never whether its objects are valid. The bridge validates and stores every object it handles, bounds volume per uplink pair and per traffic class, and never relays an object back to the peer it arrived from.

Four implementation rules carry most of the replication safety. The transport hands ingress an opaque byte string and has no ability to decode and re-encode it. Every peer, the bridge included, passes the complete validity check. Database uniqueness, not application logic, absorbs the duplicate that inevitably arrives when a live stream and a backfill both carry the same object. Finally, fan-out excludes the originating peer, which avoids a redundant round trip that content deduplication would eventually terminate but not for free.

IV. Methodology

We evaluate five questions: whether independent encoders agree on the canonical form, whether objects cross an enforced route cut, how delivery latency scales with hop count, whether malformed requests produce storage writes, and how the representation compares in size with ActivityPub. All experiments run on one physical host; §VI treats the consequences of that at length.

Correctness tests. A shared corpus of 16 positive and adversarial byte vectors is executed independently by the TypeScript, Rust, and Python implementations, each of which must agree byte-for-byte on encoding and identifier derivation and must reject every negative vector. A further 85 focused tests cover ingress (29), the outbox (3), federation (31), and simulated ISP behaviour (22). These are regression tests written by the system's authors, not an independent security audit.

Route-cut experiment. Four application nodes, two databases, and two caches are distributed across three isolated virtual network segments as in Fig. 3. Containers supply reproducible layer-2 isolation and let the harness withdraw the only configured inter-island path mid-run; nothing in the design depends on that packaging. The gate reads kernel TCP state from `/proc/net/tcp` inside the bridge to confirm which local source address each established federation connection actually left from, withdraws the exchange segment, verifies that direct probes between islands now fail and that island B does not already hold the content, submits a dependent certificate, community, and post, and polls for each exact content identifier. Nineteen assertions cover topology, source binding, relay behaviour, quota reservation, and uplink failover. Reported latencies include the application's own polling interval and are therefore upper bounds.

Hop-scaling chain. Eight nodes are peered into a line, each

with its own database, so that a node i hops from the origin is reachable by exactly one path of exactly that length. Hop count is the independent variable, which a mesh would destroy, since in a mesh every node is one hop away and only fan-out width varies. Latency is computed by subtracting the `received_at_ms` value each node writes inside the same transaction as its projection, so no polling observer sits in the measurement path; this is sound only because all nodes share the host clock, which a real deployment would have to replace with NTP discipline and an error bar. We inject 200 valid envelopes at hop 0 at a paced rate and record arrival at hop 7. The eight nodes share a single database process with one database each, which introduces contention and inflates the reported figures—the conservative direction in which to be wrong.

Signature-invalid flood. Random bytes are refused within microseconds at the parse stage and measure nothing interesting. The tool therefore lifts a genuine envelope from the node's own log and mutates a single byte inside the signed region, which invalidates the signature and changes the content identifier, so every request is novel and none can be short-circuited by deduplication. We issue 3,000 such requests at concurrency 16 and record completion rate, rejection stage, the fraction reaching signature verification, and—via the database's collection-level profiler—the resulting writes. The ordinary rate limiter stays enabled, so this is not a saturation benchmark.

Encoding comparison. We collected 80 real `Create`/`Note` activities from the public outboxes of two stock ActivityPub deployments. Because a collection hoists `@context` to the collection while an activity POSTed to a remote inbox carries its own, measuring the bare outbox item would understate delivery size; the tool refetches a standalone object from the same host and re-attaches exactly those bytes. We compare raw and gzipped activity bytes against equivalent signed envelopes, sub-

tracting authored UTF-8 text so the metric reflects what a protocol designer controls rather than how much a user typed. The comparison is deliberately generous to ActivityPub, excluding HTTP framing, TLS, per-follower fan-out, shared dictionaries, and—most significantly—authentication, since our bytes carry a 64-byte signature verifiable at rest while HTTP Signatures are transport level and vanish with the request.

V. Results

A. Correctness and convergence

All three implementations agreed on 16 of 16 canonical vectors and rejected every negative vector as specified. The focused suites additionally exercise invalid signatures, non-canonical byte layouts, replay, missing authorisation dependencies, concurrent duplicate delivery, receipts signed by the wrong key, invalid inclusion proofs, stale tree heads, and equal-size log conflicts. Sixteen vectors plainly do not exhaust protobuf's edge cases; what they establish is that three independently written implementations can be held to one byte-exact profile, which is the property a relay network depends on.

The route-cut gate passed all 19 assertions. Source-binding verification via kernel TCP state confirmed established federation connections leaving the bridge from both configured addresses, 10.90.1.30 and 10.90.2.30—an assertion no in-process harness can make, because in process there are no sockets to inspect. After direct inter-island reachability was withdrawn, island B received the certificate 0.5 s, the community 1.2 s, and the post 2.1 s after publication, in dependency order, by way of the bridge. The control matters as much as the result: the same chain measured with the exchange intact took 1.5, 1.5, and 1.9 s, so the relayed path is not merely functional but within a factor of two of the direct one. These are single preserved runs, not distributions.

That number was earned rather than observed. An earlier build passed 18 of 19, and the crossing it did complete took 407.6 s, with every outbox row still recording zero delivery attempts. The cause was serial per-peer draining with no deadline: one blackholed peer stalled delivery to every other peer in the same pass, including the bridge, whose entire purpose was to route around the partition causing the stall. Draining peers concurrently and bounding each delivery reduced the crossing to 2.1 s, a factor of 195. The defect is invisible to a two-peer harness by construction—it requires three peers, one of which becomes unreachable while the others remain up—and this is our strongest argument for evaluating federation on a deployed multi-peer testbed rather than in process.

B. Latency scaling and admission cost

All 200 envelopes traversed the eight-node chain to hop 7. End-to-end median latency was 4.125 s and the 99th percentile 6.115 s, with the per-hop distribution in Fig. 4. The curve is more informative than its endpoint. Per-hop cost is dominated by the outbox drain interval rather than by the protocol, predicting roughly $drain/2$ plus a fixed term; at a 1,000 ms drain the seven hops cost 589 ms each, implying an 89 ms fixed component, and at 250 ms they cost 196 ms each, implying 71 ms. A residual stable at 71–89 ms across a fourfold change in the queueing parameter is the protocol's genuine per-hop cost—verify, project, append, re-enqueue—and everything above it is a scheduling choice an operator makes for bandwidth and battery reasons. Contention on the shared database process and the absence of wide-area delay both mean this curve should not be read as a prediction for an Internet deployment.

The signature-invalid flood was rejected at 730 requests per second and produced zero object, index, log, or outbox writes. The negative result carries a control, since "the profiler recorded no writes" is worthless if the profiler was not running: the

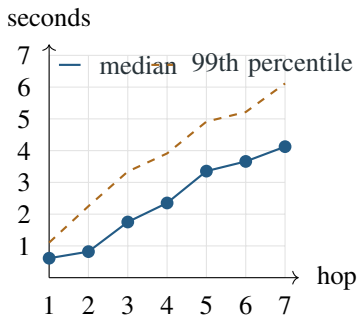

Fig. 4. Measured per-hop latency across the eight-node chain at a 1,000 ms drain interval, from in-transaction timestamps. All 200 envelopes reached hop 7.

same profile captured six read operations over the window, establishing that it was live. We report the composition of that traffic rather than the throughput figure, because the composition is what qualifies the claim. Roughly 2,700 of the 3,000 requests were shed by the per-peer rate limiter, so only about 9% reached signature verification, and the expensive rejection path—canonical parse followed by Ed25519 verify—was therefore exercised at an effective rate near 66 requests per second. The supported conclusion is narrow: for this traffic, the no-write prefix held, and it did so as the second line of defence behind a token bucket that did most of the work. The experiment says nothing about distributed floods across many peer identities, about correctly signed envelopes that are *supposed* to produce writes, or about sustained parser and database pressure.

Table I reports the encoding comparison and the costs the format incurs. Both sampled hosts emit an identical 927-byte `@context` declaration—minimum equal to maximum on each—so it is a fixed per-activity cost of the stock dialect rather than a deployment artefact, and that declaration alone is over four times the size of our entire signed forum post. Compression erases most of the apparent advantage and we lead with the compressed figure for that reason. Gzipped ActivityPub activ-

TABLE I

MEASURED REPRESENTATION SIZE AND DESIGN TRADE-OFFS AGAINST ACTIVITYPUB. THE BYTE SAMPLE IS 80 REAL CREATE/NOTE ACTIVITIES FROM TWO STOCK DEPLOYMENTS. RATIOS COMPARE GZIPPED ACTIVITYPUB AGAINST RAW SIGNED ENVELOPES, WHICH DO NOT COMPRESS.

Dimension	This work	ActivityPub comparison	Cost or boundary
Measured bytes	Raw signed envelopes: check-in 155 B, forum post 220 B, broadcast 243 B; gzip enlarges all three	Delivered JSON-LD median 4,216 B, gzipped median 1,074 B (range 786–1,400 B); `@context` alone a fixed 927 B	Ratio 4.4–6.9×; representation only—HTTP, TLS, fan-out, shared dictionaries, and AP authentication all excluded in AP's favour
Interoperability	Custom protobuf domains and RPCs	Standard actor, object, and inbox model	No Fediverse compatibility and no existing application ecosystem
Evolution	Unknown fields rejected; versioned profiles required	Flexible JSON-LD vocabulary and open extension	Harder rolling upgrades and more cross-language test surface
Operations	Generated codecs and canonical-byte tooling required	Text inspectable with ordinary web tools	Higher debugging and tooling burden on operators
Integrity profile	One exact byte string per object, content ID, author signature, and receipt	Authentication depends on deployment profile; HTTP Signatures do not survive the request	Unambiguous byte identity, but a new security profile to maintain and audit

ities ranged 786–1,400 B with a median of 1,074 B, against raw signed envelopes of 155 B, 220 B, and 243 B: a 4.4–6.9× ratio. Our envelopes do not compress at all—a 64-byte Ed25519 signature is incompressible and gzip makes all three slightly *larger*—so the honest comparison is gzipped ActivityPub against our raw bytes, not the roughly 20× ratio the uncompressed figures would suggest. Size is also the only axis on which we win. The format cannot join the existing Fediverse, requires generated codecs and unfamiliar operational tooling, rejects unknown fields so that schema evolution demands a versioned domain, and gives up the open extensibility that is JSON-LD's reason for existing.

VI. Discussion and Limitations

Testbed validity. Every experiment ran on one physical host using virtual network segments, and the resulting evidence ceiling is the paper's principal weakness. All nodes share a kernel, clock, CPU, and storage subsystem; the shared clock is what makes cross-node timestamp subtraction valid at all, and it is exactly what a real deployment does not provide. Virtual segments, static addressing, and local name resolution are far simpler than physical routers, BGP reconvergence, carrier-grade NAT, DNS failure, variable MTUs, and radio access, and attaching a dual-homed bridge to two virtual networks is trivial where a real one needs separate links, routing policy, credentials, power, and an operator willing to be identified. The route-cut experiment therefore demonstrates a change in software and routing state, not geographic latency, independent failure domains, or provider behaviour.

Access and subscriber isolation. The design assumes nodes within an island can reach one another or a shared bridge. Wi-Fi client isolation at an access point, or subscriber isolation within an ISP, can block customer-to-customer traffic while leaving every customer able to reach the provider gateway. Where that isolation holds on every surviving path, nodes can neither dial each other nor reach a bridge, federation fails entirely, and only isolated local service remains. The prototype has no non-IP fallback for this case.

Network adversary. A network operator can enumerate stable endpoints, block addresses or SNI values, fingerprint the RPC traffic, correlate timing, isolate a node, or coerce a bridge operator. Source-address binding selects an uplink; it conceals nothing. One shutdown model defeats us outright and deserves naming rather than omission: protocol whitelisting, in which domestic routing stays intact but deep packet inspection admits only sanctioned protocols. Our federation traffic is a distinctive

non-standard protocol on a distinctive port, so under whitelisting it dies on the first day regardless of how sound its signatures are. Carrying federation over ordinary HTTPS, rotating endpoints, and adopting pluggable transports are the obvious responses, and none of them is implemented or evaluated here.

System and application limits. Signatures authenticate keys, not truthfulness or safety, and a legitimate key can publish abuse or an expensive flood. Because admission cost is charged at the origin, an origin operator controls its own secrets and can mint unlimited free proofs for its own users; anti-abuse strength is therefore bounded by the policy of the weakest origin in a peer set, and per-peer quotas are the only backstop. Key theft, metadata exposure, moderation at scale, revocation recovery, storage exhaustion, and coordinated Sybil behaviour all remain open. Transport deadlines bound waiting rather than cancelling work. Consistency proofs for growing trees are specified but not implemented. Most fundamentally, a receipt evidences only what was acknowledged: an operator who declines admission outright leaves no receipt and hence no evidence, so the cheapest form of censorship—silence—is the one this construction cannot document.

Evaluation limits. The route-cut figures come from single preserved runs under one configuration, and the encoding sample is 80 activities of a single type from two deployments. We have no multi-host, wide-area, multi-day, or real-shutdown trials, and did not preserve host load, repeated-run distributions, or confidence intervals. Resource figures were taken on x86 hardware and say nothing about the single-board computers a community deployment would realistically use. Comparisons against proximity-based and voice-channel systems concern different workloads and are not a ranking.

Ethics. All tests used synthetic content on isolated private networks. A real deployment could expose bridge operators

and users to legal or physical risk, since a visible relay is by construction legible to the network that hosts it, and should not be attempted in a hostile setting without local threat analysis, informed consent, data minimisation, and community review.

VII. Conclusion

Server federation can survive a route cut when domestic IP paths remain. Canonical signed objects, a single admission path for local and federated traffic, caller-initiated recovery, and an explicitly trusted validating relay carried a dependent post across a severed exchange in 2.1 s, within a factor of two of the intact path, and the defect that stood between 407.6 s and that figure was visible only once three peers were deployed with one of them unreachable. Cross-language byte agreement and a hop-scaling chain whose per-hop cost tracks a queueing model to within 89 ms support the mechanism from two further directions. The compact representation is 4.4–6.9$\times$ smaller than gzipped ActivityPub and forfeits interoperability, extensibility, and mature tooling to get there. The work we consider most urgent is multi-host and wide-area evaluation, consistency proofs for growing logs, cancellation-aware transport, floods of validly signed traffic, redundant independent bridges, transport camouflage against protocol whitelisting, and a non-IP fallback for subscriber isolation and total blackouts.

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